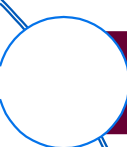
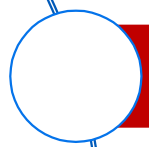


Agile development using Scrum

Agenda



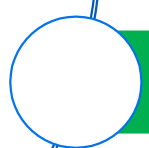
What is Agile ?



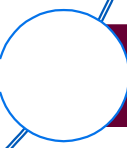
Traditional Approach vs Agile



Waterfall Model vs Agile



Agile Manifesto



Scrum

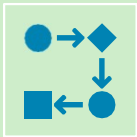
What is Agile ?



Project management and software development methodology



Empasize on flexibility, collaboration and customer centric



Focus on iterative progress through small , incremental changes rather than liner approach

Traditional Approach vs. Agile

Requirements

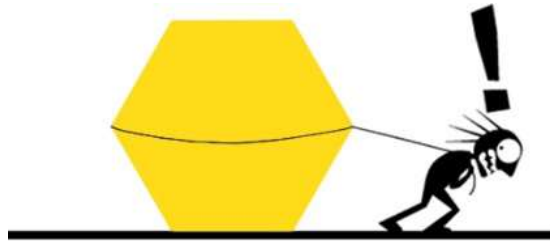
Design

Code

Test

Rather than doing all of
one thing at a time...

...Agile teams do a little of
everything all the time



Agile Vs Waterfall

Waterfall		Agile
On process	Focus	On people
By plans	Management	By changes
Linear/ sequential / long Iterations	Development Style	Incremental/iterative/short iterations
Planned at beginning	Requirements	Updated before every iteration
At the beginning and end	Customer Involvement	All the time
Minimal	Feedback	Frequent
Minimization of changes	Change Management	Accept and adapt change
As a whole at the end	Product Delivery	Frequent but partial

Agile Vs Waterfall



Why Agile ?

#01

Flexibility & Adaptability

#02

Improved Product Quality

#03

Flexibility & Adaptability

#04

Improved Product Quality

#05

Enhance Collaboration

#06

Faster Time to Market

#07

Continuous Improvement

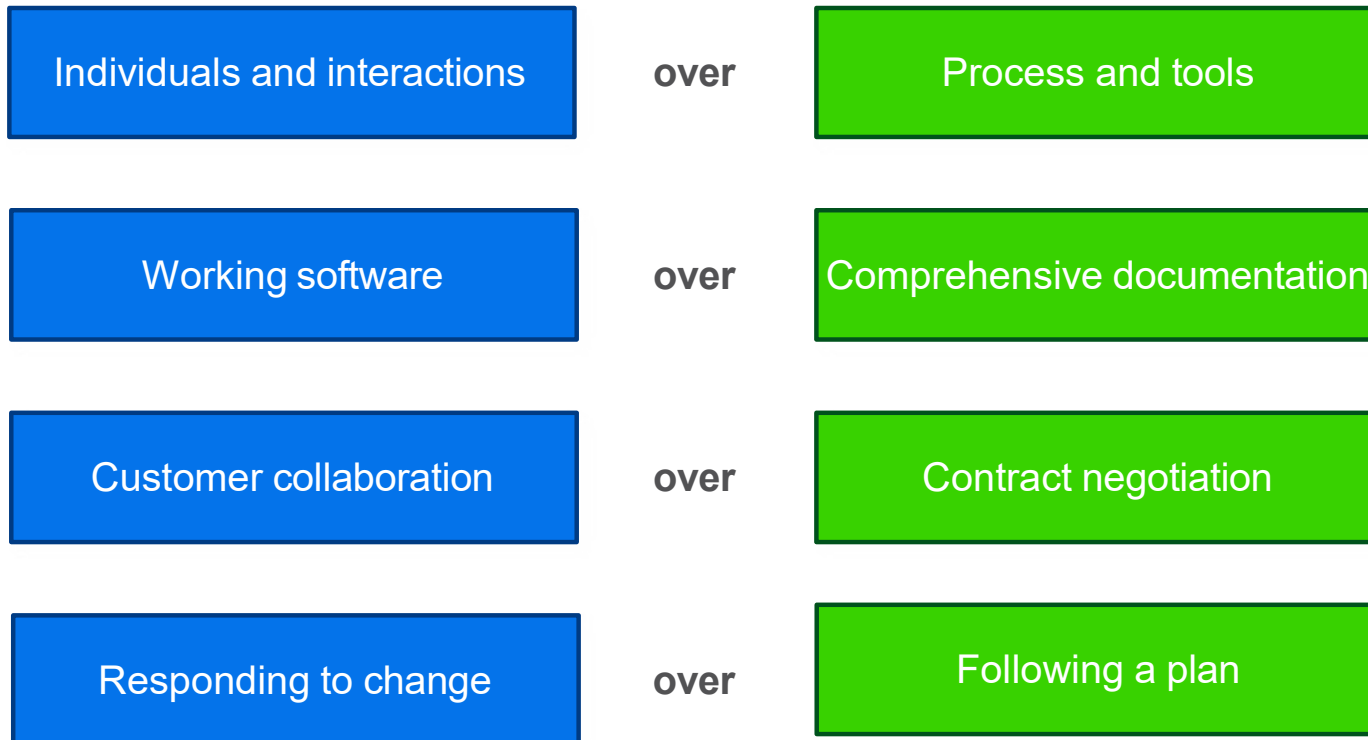
Agile Manifesto

A document that identifies 4
key values and 12 principles
to guide software
developers



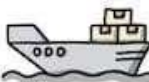
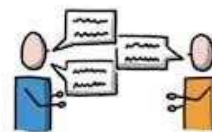
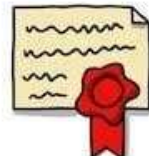
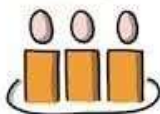
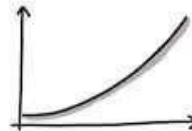
Agile Values

The 4 values of Agile Manifesto promote a software development process that focus on quality and developing products meeting customers satisfaction



Agile Principles

The 12 principles are intended to create and support a work environment that is focused on the customer and customer satisfaction

Satisfy the customer through early and continuous delivery of valuable software. 	12 Agile Principles @OlgaHeismann		Business people and developers must work together. 
	 Welcome changing requirements, even late in development.	 Deliver working software frequently.	
Build projects around motivated individuals. Give them the support they need. Trust them. 	 The most efficient and effective method of conveying information is face-to-face conversation.	Working software is the primary measure of progress. 	 The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
Continuous attention to technical excellence and good design. 	 Simplicity — the art of maximizing the amount of work not done — is essential.	The best architectures, requirements, and designs emerge from self-organizing teams. 	The team reflects on how to become more effective and adjusts its behavior accordingly. 

Agile Framework

Agile methodology supports few frameworks:

1. Crystal
- 2. Scrum**
3. Kanban



Scrum is a lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems.

- From The Scrum Guide, Nov 2020



Scrum Values

1

Commitment

2

Focus

3

Openness

4

Respect

5

Courage



Roles (3)

1. Product Owner
2. Scrum Master
3. Developers



Artifacts (3)

1. Product backlog
2. Sprint backlog
3. Increment



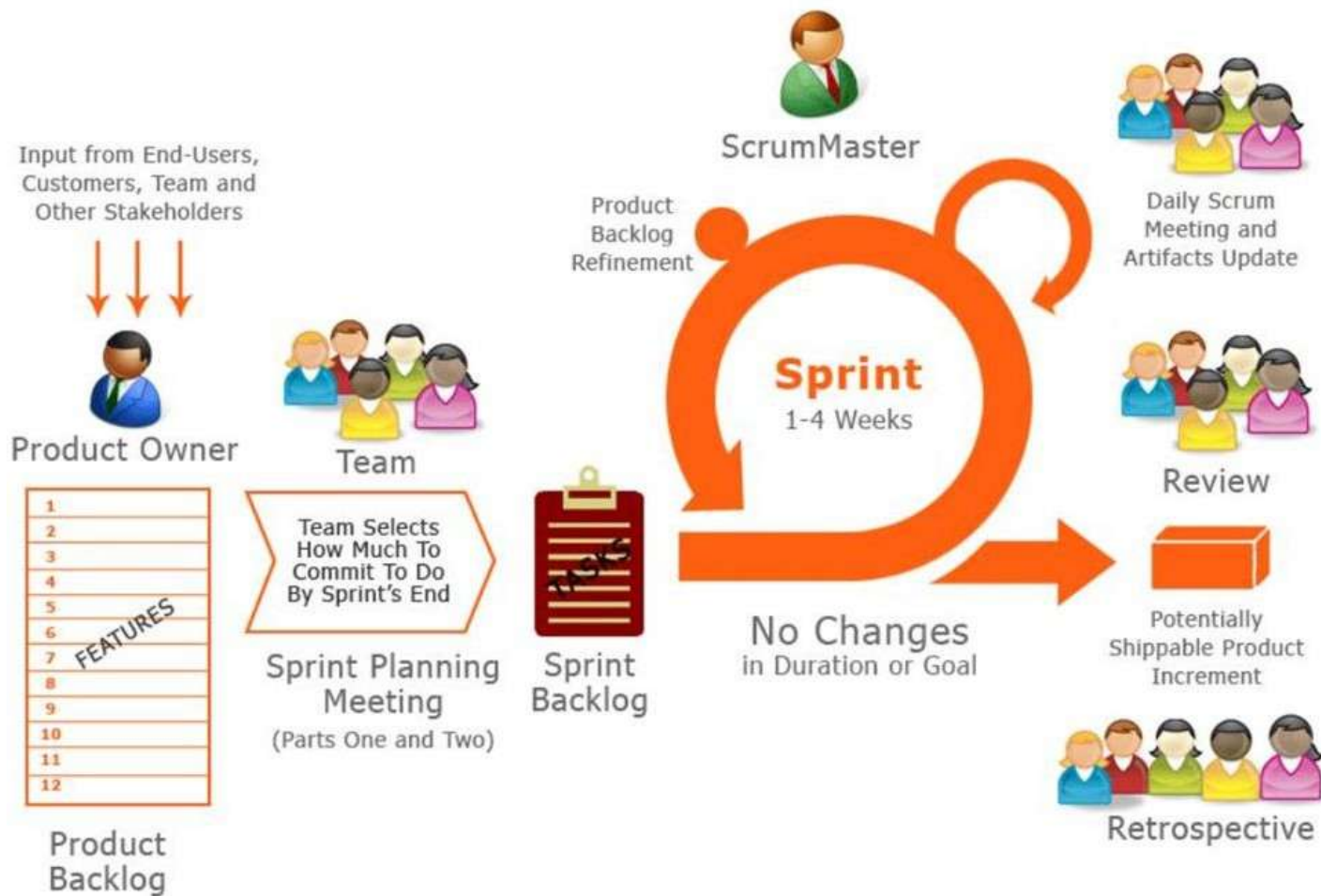
Events (4)

1. Sprint Planning
2. Daily Stand Up
3. Sprint Review
4. Sprint Retrospective



Scrum Overview

Scrum Framework



A Scrum Team



**One
Product Owner**



**One
Scrum Master**



Developers

Product Owner (One person, not a committee)

- The Product Owner is accountable for maximizing the value of the product resulting from the work of the Scrum Team, which includes:
 - Creates the Product Vision and drives product success
 - Defines the feature of the product
 - Prioritize features according to market value
 - Decide on release date and content
 - Accept or reject work results
- The Product Owner is also accountable for effective Product Backlog management, which includes:
 - Developing and communicating the Product Goal
 - Creating and clearly communicating Product Backlog items
 - Ordering Product Backlog items
 - Ensuring that the Product Backlog is transparent, visible and understood



Scrum Master

- The Scrum Master is accountable for establishing Scrum as defined in the Scrum Guide.
- The Scrum Master is accountable for the Scrum Team's effectiveness.
- Scrum Masters are true leaders who serve the Scrum Team and the larger organization.
- The Scrum Master serves the Scrum Team in several ways, including:
 - Coaching the team members in self-management and cross-functionality
 - Helping the Scrum Team focus on creating high-value Increments that meet the Definition of Done
 - Causing the removal of impediments to the Scrum Team's progress
 - Ensuring that all Scrum events take place and are positive, productive, and kept within the timebox.

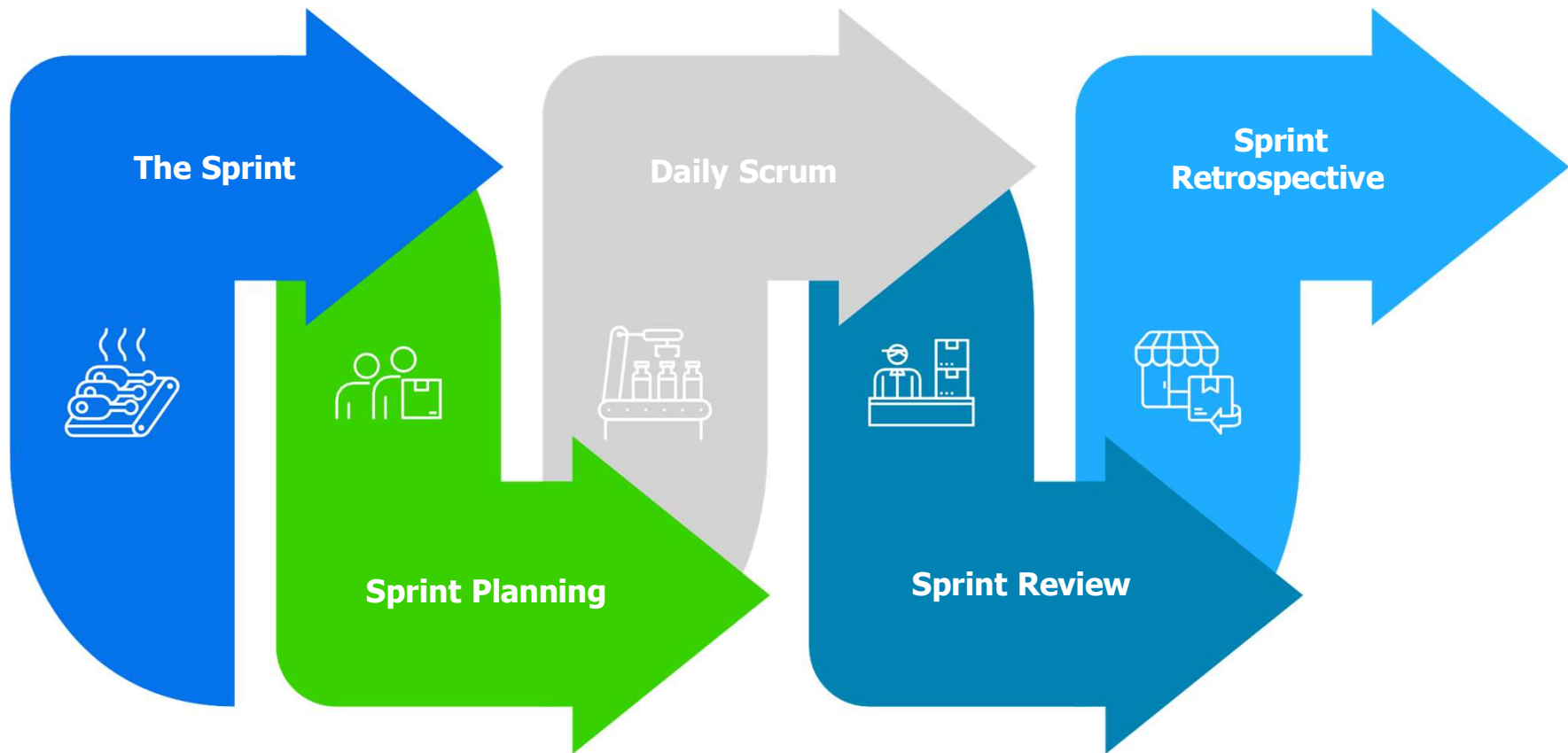


Developers

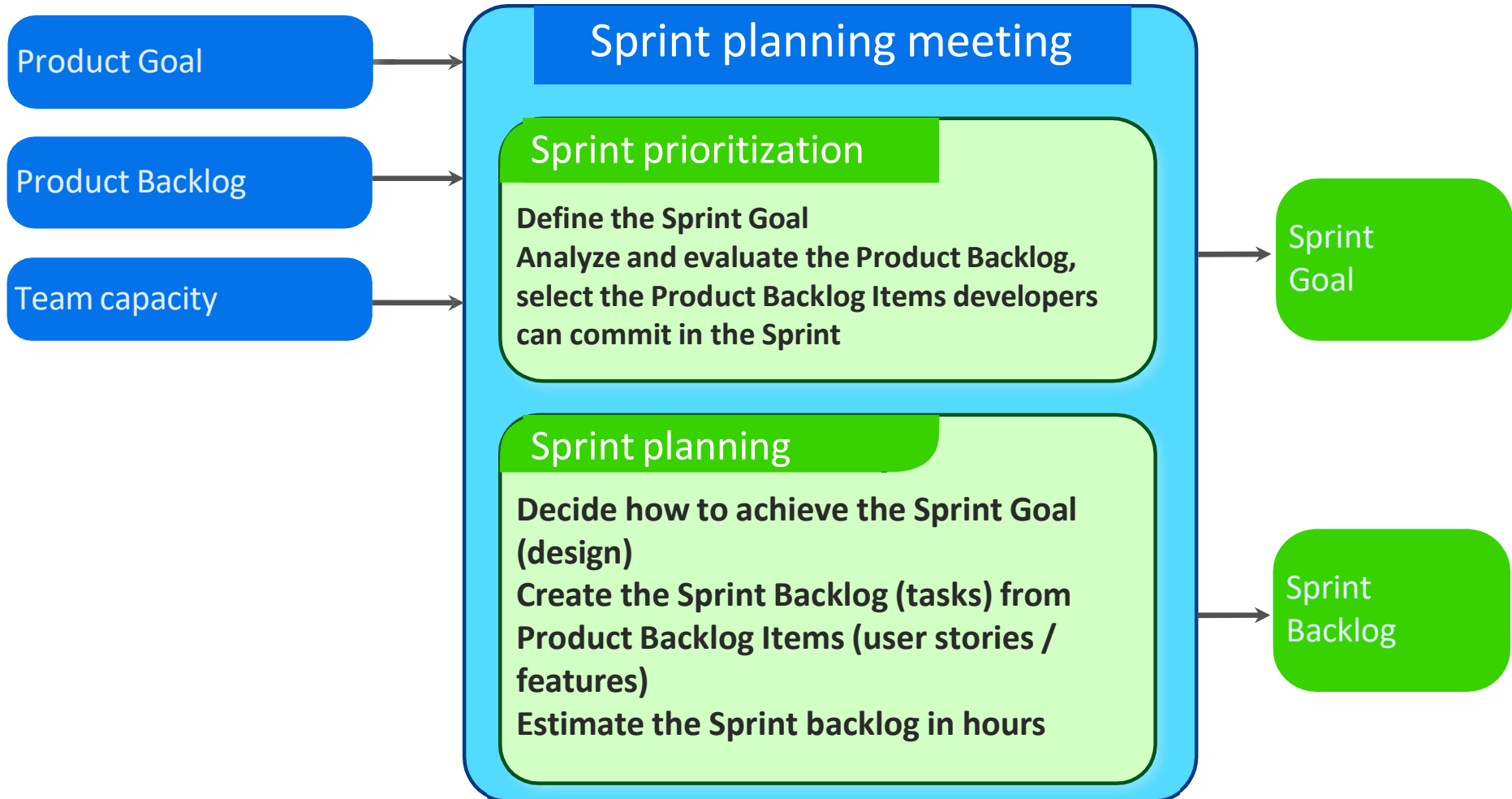


- Developers are the people in the Scrum Team that are committed to creating any aspect of a usable Increment each Sprint.
- Typically 5-9 people
- Cross-functional:
 - Programmers, testers, user experience designers, etc.
- Teams are self-managed
- The Developers are always accountable for:
 - Creating a plan for the Sprint, the Sprint Backlog
 - Instilling quality by adhering to a Definition of Done
 - Adapting their plan each day toward the Sprint Goal

Scrum Events



Sprint Planning



Sprint Goal

- The Sprint Goal is the single objective for the Sprint. A commitment by the Developers.

Examples Sprint Goals

Deliver a minimal set of administration features

Sprint

01

The Sprint is a container for all other Scrum events

02

Each Sprint may be considered a mini-project

03

Typical duration is 2–4 weeks or a calendar month at most

04

A new Sprint starts immediately after the conclusion of the previous Sprint.

05

Product is designed, coded, and tested during the sprint

06

In a Sprint, each time-boxed event is designed to provide a inspect and adapt opportunity.

07

There is always tangible input and output for each event.

08

At the end of a sprint, a “Done”, useable, and potentially releasable product Increment is created

09

The Sprint provides stability in a world of unpredictability.

Daily Scrum

- Purpose: inspect progress toward the Sprint Goal and adapt the Sprint Backlog, adjusting the upcoming planned work.
- Parameters
 - Daily
 - 15-minutes
 - Stand-up
 - Same time, same place
- Not for problem solving
 - Only team members, ScrumMaster, Product Owner, can talk

(If the Product Owner or Scrum Master are actively working in the Sprint Backlog, they participate as Developers)
- Helps avoid other unnecessary meetings, improve communication, promote quick decision making



Sample: Everyone answers 3 Questions

What did I get DONE yesterday?

What will I get DONE today?

Is anything in my way?

- It's broadcast in front of peers for self-management

Sprint Review

- Team presents what is accomplished during the Sprint to key stakeholders
- Inspect and adapt on the Product Backlog
- Product Owner accept “Done” work and reject “un-done” work
- Typically takes the form of a demo of new features or underlying architecture
 - A maximum of 4 hours for a one-month Sprint.
 - For shorter Sprints, the event is usually shorter.
 - No slides
- Whole Scrum team participates
- Invite the world

Sprint Retrospective

- Inspect and adapt on how we work, plan ways to increase quality and effectiveness
- Periodically takes a look at what is and is not working
- Typically 60 min, max 3 hours for a one-month Sprint.
- Done after every sprint
- Whole team participates
 - Scrum Master
 - Product owner
 - Developers
- Improvement action plan for next sprint

Start doing

Stop doing

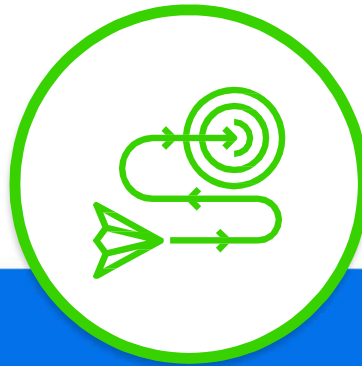
Continue doing

Scrum Artifacts

Product backlog



Sprint backlog



Increment



Each Artifact contains a commitment to ensure it provides information that enhances transparency and focus against which progress can be measured:

- 1) For the Product Backlog it is the Product Goal;
- 2) For the Sprint Backlog it is the Sprint Goal;
- 3) For the Increment it is the Definition of Done;

Product Backlog

- An ordered list of everything that is known to be needed in the product.
- It lists all features, functions, requirements.
- Ideally expressed such that each item has value to the users or customers of the product
- Prioritized by the Product Owner
- Reprioritized at the start of each Sprint

Commitment: Product Goal

- The Product Goal describes a future state of the product which can serve as a target for the Scrum Team to plan against.
- The Product Goal is in the Product Backlog

Sprint Backlog

- The Sprint Backlog is composed of the Sprint Goal (**Why**), the set of Product Backlog items selected for the Sprint (**What**), as well as an actionable plan for delivering the Increment (**How**).
- Individuals sign up for work of their own choosing
 - Work is never assigned
 - Sign up for new work after one work is done
- Team tracks remaining work of the Sprint, daily.
- Any team member can add, delete or change the sprint backlog
- A highly visible, real-time picture of the work, task board is a common tool.

Commitment: Sprint Goal

- The Sprint Goal is the single objective for the Sprint. A commitment by the Developers.
- The Sprint Goal is created during the Sprint Planning event and added to the Sprint Backlog.
- The Sprint Goal is :
 - Potentially
 - Shippable
 - Product
 - Increment

Increment

- An Increment is a sum of all the Product Backlog items completed
- The sum of the Increments is presented at the Sprint Review.
- Work cannot be considered part of an Increment unless it meets the Definition of Done.
- Must be in **useable condition** regardless whether the Product Owner decides to release

Commitment: Definition of Done

- The Definition of Done is a formal description of the state of the Increment when it meets the quality measures required for the product.
- The moment a Product Backlog item meets the Definition of Done, an increment is born.
- The Definition of Done creates transparency by providing team a shared understanding of what work was completed as part of the Increment.

Thank You